

Day 71 – Integrated Network Services & Enterprise Network Design

Overview

Day 71 was the first networking lab that focused on **how all previously learned technologies work together as a complete enterprise network** rather than studying individual technologies separately.

In previous labs, each technology was introduced independently:

- Day 65 – RIP Dynamic Routing
- Day 66 – DHCP
- Day 67 – DNS Concepts
- Day 68 – NAT
- Day 69 – ACL Security
- Day 70 – VLAN Segmentation

Day 71 combined these concepts into a single enterprise network architecture and demonstrated how devices, routers, security controls, and network services interact during everyday communication.

Objectives

By the end of this lab, I was able to:

- Understand how multiple network services work together.
- Verify DHCP functionality.
- Verify RIP routing functionality.
- Verify NAT functionality.
- Verify ACL security policies.
- Understand the role of VLANs in network segmentation.
- Analyze traffic flow through an enterprise network.
- Connect networking concepts to cloud architecture.

Enterprise Network Overview

The existing topology from previous labs was maintained.

PC0 PC1 PC2

|||

SW0 SW1 SW2

|||

R0 ----- R1 ----- R2 ----- Router3

|

Server0

,

Services included:

✓ RIP

✓ DHCP

✓ NAT

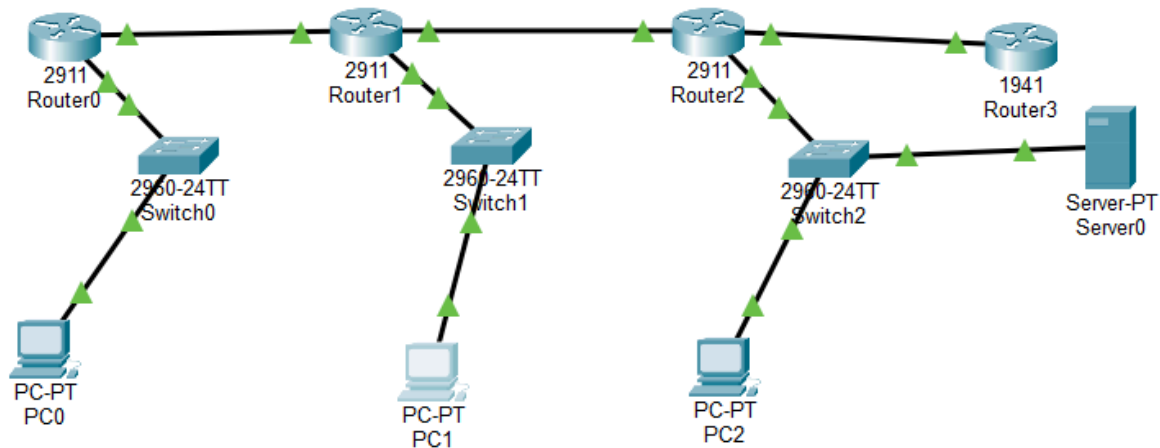
✓ ACLs

✓ VLAN Concepts

✓ Server Infrastructure

✓ ISP Simulation

Screenshot #1 – Complete Enterprise Topology



Highlight

- Three LANs
 - Three internal routers
 - ISP Router
 - Server0
 - Multiple network services running simultaneously
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How The Network Works

Day 71 focused on understanding what happens when a user powers on a computer and accesses a resource.

Step 1 – DHCP

When a client device starts:

PC0 boots.

The PC initially has:

No IP address.

The PC sends:

DHCP Discover

A router acting as a DHCP server responds.

The client receives:

IP Address

Subnet Mask

Default Gateway

DNS Information

Day 66 functionality.

Step 2 – DNS

When a user accesses a hostname:

server.local

DNS translates:

server.local

into:

192.168.30.100

Day 67 functionality.

Step 3 – RIP

The destination network must be located.

Routers check:

Routing Tables

RIP provides:

Remote Network Information

Routers learn where to forward traffic.

Day 65 functionality.

Step 4 – ACL Evaluation

Before forwarding traffic:

Router

↓

ACL Check

The router determines:

Permit?

or

Deny?

Day 69 functionality.

Step 5 – NAT

If traffic is leaving the private network:

192.168.x.x

NAT translates:

Private Address

↓

Public Address

allowing Internet access.

Day 68 functionality.

Service Verification

DHCP Verification

Command:

show ip dhcp binding

Verified that DHCP clients successfully received addresses.

Screenshot #2 – DHCP Verification

```
Router>show ip dhcp binding
IP address      Client-ID/      Lease expiration    Type
                Hardware address
192.168.20.2    0090.219C.51CC  --                  Automatic
Router>
```

Highlight

Automatic Address Assignments

RIP Verification

Command:

show ip route

..

Verified:

R Routes

indicating RIP learned networks.

Examples:

R 192.168.20.0

R 192.168.30.0

R 209.165.200.0

Screenshot #3 – RIP Routing Table

```
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/1
R       10.0.0.4/30 [120/1] via 10.0.0.2, 00:00:26, GigabitEthernet0/1
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/26 is directly connected, GigabitEthernet0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0
R       192.168.20.0/24 [120/1] via 10.0.0.2, 00:00:26, GigabitEthernet0/1
R       192.168.30.0/24 [120/2] via 10.0.0.2, 00:00:26, GigabitEthernet0/1
R       209.165.200.0/24 [120/2] via 10.0.0.2, 00:00:26, GigabitEthernet0/1
_
```

Highlight

R Routes

,

for learned networks.

NAT Verification

Command:

show ip nat translations

Traffic was generated using:

ping 209.165.200.226

NAT translated:

Private Addresses

into:

Public Addresses

through Router2.

ACL Verification

The ACL configured during Day 69 remained active.

Testing:

PC0

ping 192.168.30.2

Result:

Blocked

because LAN10 was denied access to LAN30.

Testing:

PC1

ping 192.168.30.2

Result:

Successful

because LAN20 remained permitted.

Screenshot #5 – ACL Enforcement

```
C:\>ping 192.168.30.2

Pinging 192.168.30.2 with 32 bytes of data:

Reply from 10.0.0.6: Destination host unreachable.
Reply from 10.0.0.6: Destination host unreachable.
Reply from 10.0.0.6: Destination host unreachable.
Reply from 10.0.0.6: Destination host unreachable.

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Highlight

LAN10 Blocked

LAN20 Allowed

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.2

Pinging 192.168.30.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.2: bytes=32 time<1ms TTL=126
Reply from 192.168.30.2: bytes=32 time<1ms TTL=126
Reply from 192.168.30.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

VLAN Integration

Day 70 introduced VLAN segmentation.

VLANs provided:

Logical Separation

between groups of users.

Examples:

HR VLAN

Sales VLAN

This concept demonstrates how organizations separate users while still using shared infrastructure.

Enterprise Traffic Flow

A complete communication sequence now looks like:

PC0
↓
DHCP
↓
Receives Address
↓
DNS
↓
Resolves Name
↓
RIP
↓
Finds Route
↓
ACL
↓
Security Check
↓
NAT
↓
Internet Access
..

This is the first fully integrated enterprise workflow built during Phase 4.

What Day 71 Demonstrated

Rather than focusing on one technology, this lab demonstrated how:

DHCP
DNS
RIP
ACLs
NAT
VLANs

work together.

The focus shifted from:

What does this technology do?

to:

How do all these technologies interact?

Cloud Engineering Connection

The services used throughout Phase 4 directly align with cloud architecture concepts.

Networking Technology	Cloud Equivalent
DHCP	AWS DHCP Options
DNS	Route 53
RIP	Route Tables
ACLs	Security Groups / NACLs
NAT	NAT Gateway
VLANs	Subnet Segmentation
ISP Router	Internet Gateway Concept

The entire Phase 4 network now resembles a simplified cloud environment.

What I Learned

- Enterprise networks rely on multiple services simultaneously.
- DHCP provides addressing information.

- DNS provides name resolution.
 - RIP provides routing knowledge.
 - ACLs enforce security.
 - NAT provides Internet access.
 - VLANs provide network segmentation.
 - Network technologies do not operate independently.
 - Cloud networking concepts are built on these same foundations.
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Reflection

Day 71 served as the integration point for all networking technologies learned throughout Phase 4. Rather than introducing a new protocol or service, the lab demonstrated how DHCP, DNS, RIP, ACLs, NAT, and VLAN concepts work together inside a single enterprise network. This helped shift my understanding from individual networking technologies toward complete network architecture and provided a stronger connection between traditional networking concepts and modern cloud environments.
